



SENFENG

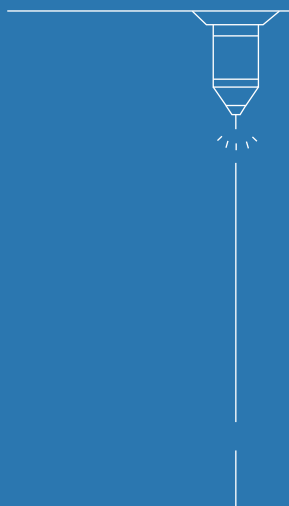
HOW TO SELECT YOUR LASER CLEANING MACHINES

www.senfenglaser.com



**PROVIDE GLOBAL CUSTOMERS WITH METAL
FORMING AUTOMATION SOLUTIONS**

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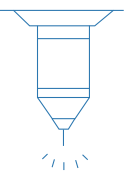
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About Laser Cleaning

Laser cleaning is a new technology based on the interaction between laser and materials, which can make high-density laser beams shine on workpiece surface until dirt, rust or coating instantly vaporizes or peels off to clean the workpiece. Unlike traditional mechanical, chemical and ultrasonic cleaning, it can do without any CFC-type organic solvents that may damage the ozone sphere. No pollution or noise, nor harm to human body and environment. This is a green cleaning technology.

Laser cleaning is also superior to mechanical friction, chemical corrosion, liquid-solid impact cleaning and high-frequency ultrasonic cleaning in terms of ease of operation, automated integration and necessity of chemical reagents. Besides high cleanliness and precision, laser cleaning can also work on cambered workpiece at high efficiency, without any harm to worker's body and substrate surface. In doing so, traditional cleaning will be a thing of past.

Laser cleaning has been widely applied in fields such as aerospace, automotive production, large-scale integrated circuit, military project, hi-end machine tool, rail transit, eco-friendly cultural relic repair and premium mold. It has brought revolutionary changes to these areas and offered various solutions for resins, paints, grease, rust, coatings, platings, oxidation layer and other contaminants on the surface of workpieces in different industries.





SENFENG Laser Welding & Cleaning Technical Center

SENFENG, one of the earliest companies founding laser welding & cleaning centers in China, has stood at the forefront of science and technology and clung on to customer-oriented demands to carry out automatic application tests and feasibility study on various materials, cleaning modes and different fields. Our cooperative partners will get free proofing, cleaning process verification and many other cleaning solutions. We are committed to being an internationally famous laboratory and guiding laser cleaning field to move towards innovative development.

At present, there are two kinds of products in SENFENG, namely, handheld and automatic laser cleaning categories, including dozens of different machines. They are currently used for aerospace, automotive production, large-scale integrated circuit, military project, hi-end machine tool, rail transit, eco-friendly cultural relic repair and premium mold. Our aim is to make laser cleaning machines be applied in industrial production.



SENFENG Laser Welding & Cleaning Testing Laboratory

In this laboratory, there are tensile machine, metallographic microscope, metallographic sample polishing machine, metallographic sample pointing machine, friction wear testing machine, Vickers hardness tester, temperature detector, surface roughometer and digital microscope used to support laser welding & cleaning testing demands. We also issue professional testing reports and provide all new and regular customers with highly stable laser cleaning machines that enable excellent cleaning effects.



Pulsed
Laser Cleaning



VS



Continuous-wave
Laser Cleaning



Pulsed Laser Cleaning



Laser outputs in a form of nanosecond pulses, delivering high energy per pulse. Its instant release of energy will make stains absorb the energy in a very short time, quickly expanding and fracturing, while generating thermal shock or impact waves that shake off or strip the stains.

The pulsed laser energy is concentrated on stain layer, resulting in a minimal heat-affected zone (usually only at micron level), which makes negligible thermal damage to the substrate and protects it at the greatest extent.

Fit for precise, sensitive or substrate-protective applications, such as:

1. Precision components (electronic elements, circuit board, mold): avoid thermal damage
2. Heat-sensitive materials (plastics, rubber, composite materials): prevent substrate from melting or deforming
3. Cultural relics and artworks: preserve original surface textures and reject excessive removal
4. Complex structures (such as gear tooth surface, inner wall of micro-hole): take precise control of cleaning range without damaging details

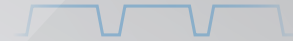
VS

Classify by Action Mechanism of Laser Energy

Classify by Thermal Effect and Harm to Substrate

Classify by Applicable Scenes

Continuous-wave Laser Cleaning



Its laser beams are continuous and stable. Such sustained thermal conduction causes stains (such as rust layers and coatings) to absorb heat, then, melt and vaporize, detaching from the substrate.

Continuous laser heats are easy to conduct to the substrate, with large heat-affected zone, which may lead to deformation, oxidation, annealing or performance degradation (such as a decrease in metal hardness) of the substrate, especially for heat-sensitive ones like plastics, rubber and precision metals.

Suit for cleaning large-area rough surfaces with strong heat resistance, where cleaning is fast but requirements on precision and substrate protection are relatively low, such as large-scale steel structure rust removal, thick paint stripping, casting parts sand cleaning, ship rust removal and coating removal from steel plates.



Laser Cleaning Machines

■ Pulsed Laser Cleaning Machines

Handheld Air-cooling Pulsed Laser Cleaning Machine

SF-200HC / SF-300HC / SF-500HC

High-power Handheld Water-cooling Pulsed Laser Cleaning Machine

SF-1000HC

■ Continuous-wave Laser Cleaning Machines

Handheld Air-cooling Continuous-wave Laser Cleaning Machine

SF-1200HC

Handheld Water-cooling Continuous-wave Laser Cleaning Machine

SF-1500HC / SF-2000HC / SF-3000HC

Medium-power Handheld Water-cooling Continuous-wave Laser Cleaning Machine

SF-6000HC

■ Laser Cleaning Solutions

Steel Plate Laser Cleaning Machine

Tyre Inner Liner Laser Cleaning Machine

Pressure Vessel Laser Cleaning Machine

Pulsed Laser Cleaning Machines

SF-HC Handheld Air-cooling Pulsed Laser Cleaning Machine

► Small in size but no worry about cleaning

This new high-tech surface treatment machine is outfitted with special air-cooling pulsed lasers. It is small in size but durable, easy to install and operate, making >8h continuous cleaning possible. Due to IO avionics connector, cleaning automation will be near at hand.

The machine, featuring fast cleaning and no harm to substrate, has been widely used in mold, automotive manufacturing and repair, ship maintenance, as well as aircraft restoration.

It can accurately remove contaminants such as rust, paint and oxidation layer, adapting to various curved and irregular surfaces. Such non-contact, eco-friendly cleaning mode, without additional chemical agents, will last for 100,000h in total.



8 Competitive Strengths

- Easy to operate and no harm to substrate
- IO interface with automation devices
- Small in size and light in weight
- Special air-cooling lasers and laser cleaning head
- No worry about high temperature due to air cooling
- Offer accurate cleaning at high quality
- 8 kinds of cleaning modes
- 10 safety alarm systems

* Machine appearance, function description and technical parameters on this page are derived from SENFENG's Technical R&D Center based on our self-developed machines. For reference only, in kind prevail!

SF-HC

SF-HC Handheld Air-cooling Pulsed Laser Cleaning Machine

Model	SF200HC	SF300HC	SF500HC
Laser power	0.2kW	0.3kW	0.5kW
Max pulsed laser energy	2mJ		15mJ
Wavelength	1064nm		
Laser output instability	<5%		
Net weight	25kg		30kg
Gross weight	35kg		40kg
Net size	665*334.5*574mm		
Packaging size	678*420*791mm(length, width and height are internal size)		
Cleaning ability	Remove oil, rust, paint and so on.		
Total length of cleaning gun line	5m		10m
Weight of laser cleaning head	0.7kg		1.2kg
Oscillatory laser spot	Straight-line, oval, splayed, rectangular, helical, petaloid, wavy, grid-shaped style 		
Cleaning width	≤100mm		≤120mm
Mode of cooling	Air cooling		
Total power	1kW	1.5kW	2.5kW
Languages	15 kinds of languages like Chinese, English, Japanese, Korean, French, German, Italian		
Power supply	Single-phase 220V 50/60HZ		

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SF300HC



SF200HC/SF300HC
Laser Cleaning Head



SF500HC



SF500HC
Laser Cleaning Head

Pulsed Laser Cleaning Machines

SF1000HC High-power Handheld Water-cooling Pulsed Laser Cleaning Machine

► Easy to operate and no harm to substrate

The high-power water-cooling pulsed laser cleaning machine makes full use of its pulsed laser modules, well-structured water cooling system and special laser cleaning head to speed up its cleaning and accept the thicker coating layer.

It will remove rust, paint and oxidation layer from workpiece surface without causing damages. Thanks to its >8h continuous cleaning and over 100,000h lifespan in total, it has been widely used in mold, automotive manufacturing and repair, ship maintenance and aircraft restoration.

Moreover, no additional chemical reagents are needed, with no media or dust. When powered on, no cleaning water is required to remove paint, stains, dirt, rust and coatings from workpiece surface.



4 Competitive Strengths

- Water-cooling pulsed lasers and dual-oscillating laser cleaning head
- 8 kinds of cleaning modes available
- Easy to store laser cleaning gun head and wirecables
- 10 safety alarm systems

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SF-HC

SF1000HC High-power Handheld Water-cooling Pulsed Laser Cleaning Machine

Model	SF1000HC
Max pulsed laser energy	15mJ
Laser power	1kW
Wavelength	1064nm
Laser output instability	<5%
Net weight	200kg
Gross weight	280kg
Net size	1010*745*1050mm
Packaging size	1400*870*1265mm
Cleaning ability	Remove oil, rust, paint and so on.
Total length of cleaning gun line	10m
Weight of laser cleaning head	1.5kg
Oscillatory laser spot	Straight-line, oval, splayed, rectangular, helical, petaloid, wavy, grid-shaped style 
Cleaning width	≤120mm
Mode of cooling	Water cooling
Total power	6kW
Languages	15 kinds of languages like Chinese, English, Japanese, Korean, French, German, Italian
Power supply	Single-phase 220V 50/60HZ

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SF1000HC Laser Cleaning Head

Continuous-wave Laser Cleaning Machines

SF1200HC Handheld Air-cooling Continuous-wave Laser Cleaning Machine

► Small in size but no worry about cleaning

This new high-tech surface treatment machine is outfitted with special air-cooling pulsed lasers. It is small in size but durable, easy to install and operate, making >8h continuous cleaning possible. Due to IO avionics connector, cleaning automation will be near at hand.

Its cleaning width can reach up to 300mm, without the need for chemical reagents when removing rust, paint and oxidation layer from workpiece surface. Given this, it has been widely used in sheet metal, robot production, automotive manufacturing and repair, as well as ship maintenance. Moreover, it can serve for over 100,000h in total.



4 Competitive Strengths

- Professional continuous-wave laser cleaning head
- No worry about high temperature due to air cooling
- Small in size and light in weight
- Offer accurate cleaning at high quality

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SF-HC

SF1200HC Handheld Air-cooling Continuous-wave Laser Cleaning Machine

Model	SF1200HC
Adjustable range of laser power	10-100%
Wavelength	1080±10nm
Fiber core diameter	20um
Net weight	36kg
Gross weight	45kg
Net size	665*334.5*574mm
Packaging size	678*420*791mm (length, width and height are internal size)
Cleaning ability	Remove oil, rust, paint, oxidation layer and so on.
Total length of cleaning gun line	10m
Collimating lens: focusing lens	F60:F800
Cleaning width	≤300mm
Weight of laser cleaning head	0.7kg
Total power	4.5kW
Languages	19 kinds of languages like Chinese, English, Japanese, Korean, Russian, French, German
Power supply	Single-phase 220V 50/60HZ

Optional Accessories

Model of laser cleaning head	FWH30-DC30B dual-oscillating continuous-wave laser cleaning head
Cleaning width	≤300mm
Oscillatory laser spot	Swing laser spot: linear, vertical linear, elliptical, heliciform, small dot-like, sawtooth waveform, petal, concentric circular, large grid-like type, petaloid, wavy, grid-shaped style 

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SF1200HC Laser Cleaning Head

Continuous-wave Laser Cleaning Machines

SF-HC Handheld Water-cooling Continuous-wave Laser Cleaning Machine

► Easy to operate for fast cleaning

The high-power water-cooling continuous-wave laser cleaning machine makes full use of its continuous-wave laser modules, well-structured water cooling system and special laser cleaning head to raise its cleaning width to 300mm at most.

There is no need for chemical reagents when removing rust, paint and oxidation layer from workpiece surface. Given this, it has been widely used in sheet metal, robot production, automotive manufacturing and repair, as well as ship maintenance. Moreover, it can work for >8h continuously and serve for over 100,000h in total.

Moreover, there is no media or dust. When powered on, no cleaning water is required to remove paint, stains, dirt, rust and coatings from workpiece surface.



4 Competitive Strengths

- Adopt water chiller with fast cooling and supporting >8h continuous cleaning

- Laser cleaning head, driven by digital servo motor, features high anti-interference capability

- Enable 10-20m ultra-long-distance cleaning

- Protective and reflective lens under real-time monitoring and sensitive temperature control

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SF-HC

SF-HC Handheld Water-cooling Continuous-wave Laser Cleaning Machine

Model	SF1500HC	SF2000HC	SF3000HC
Laser power	1.5kW	2kW	3kW
Wavelength	1080±10nm		
Fiber core diameter	20um	25um	50um
Net weight	165kg		210kg
Gross weight	230kg		290kg
Net size	986*489.5*900mm		1035*570*900mm
Packaging size	1400*700*1120mm		
Cleaning ability	Remove oil, rust, paint, oxidation layer and so on.		
Total length of cleaning gun line	10m		15m
Weight of laser cleaning head	0.7kg		
Cleaning width	≤300mm		
Total power	9kW	12kW	15kW
Languages	19 kinds of languages like Chinese, English, Japanese, Korean, Russian, French, German		
Power supply	Single-phase 220V 50/60HZ for 2000W and below, or three-phase 380V 50/60HZ for 3000W		

Optional Accessories

Model of laser cleaning head	FWH30-DC30B dual-oscillating continuous-wave laser cleaning head
Cleaning width	≤300mm
Oscillatory laser spot	Swing laser spot: linear, vertical linear, elliptical, heliciform, small dot-like, sawtooth waveform, petal, concentric circular, large grid-like type, petaloid, wavy, grid-shaped style 

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SF1500HC/SF2000HC/SF3000HC
Laser Cleaning Head

Continuous-wave Laser Cleaning Machines

SF-HC Medium-power Handheld Water-cooling Continuous-wave Laser Cleaning Machine

► Easy to operate for fast cleaning

The medium-power water-cooling continuous-wave laser cleaning machine makes full use of its 6kW continuous-wave laser modules, well-structured water cooling system and special laser cleaning head to raise its cleaning width to 300mm at most.

There is no need for chemical reagents when removing rust, paint and oxidation layer from workpiece surface. Given this, it has been widely used in sheet metal, robot production, automotive manufacturing and repair, as well as ship maintenance. Moreover, it can work for >8h continuously and serve for over 100,000h in total.

There is no media or dust. When powered on, no cleaning water is required to remove paint, stains, dirt, rust and coatings from workpiece surface.



4 Competitive Strengths

- 6kW laser cleaning head
- APP remote control
- Take real-time monitoring and offer 6 warning lights
- Safety lock and other protective switches available

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SF-HC

SF-HC Medium-power Handheld Water-cooling Continuous-wave Laser Cleaning Machine

Model	SF6000HC
Laser power	6kW
Wavelength	1080±10nm
Fiber core diameter	50um
Net weight	235kg
Gross weight	350kg
Net size	1418*730*1219mm
Packaging size	1600*870*1450mm
Cleaning ability	Remove oil, rust, paint, oxidation layer and so on.
Total length of cleaning gun line	20m
Collimating lens: focusing lens	F75:F900
Cleaning width	200-500mm (standard width≤300mm)
Weight of laser cleaning head	2.2kg
Total power	20kW
Languages	30 kinds of languages like Chinese, English, Korean, Russian, Japanese, French, German
Power supply	Three-phase 380V 50/60HZ



SF6000HC Laser Cleaning Head

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Laser Cleaning Solutions

Steel Plate Laser Cleaning Machine

► Precise cleaning at large format

It consists of laser generator, laser cleaning head, water chiller, dust collector, steel plate conveyor line, support frame, dust collection hood, protective sheet metal and more, designed for cleaning steel plates of various sizes (length: >800mm, width: 200-1300mm, thickness: 5-50mm). Customization is flexibly permitted. In addition, it may also be put in an automatic production line to collect and dispose cleaning smoke and dust synchronously, keeping ambient environment clean.

4 Competitive Strengths

- Give stable laser beam output
- One-click intelligent control
- Fit for cleaning large-sized workpiece
- 8 kinds of cleaning modes available



Technical Parameters

Laser power	1.5-20kW	
Cleaning mode	Laser, non-contact	
Cleaning materials	Remove oil, rust, paint, etc.	
Cleaning parameters	Oxidation layer: $\leq 100\mu\text{m}$	14-25m ² /h
	Paint, coating layer: $\leq 120\mu\text{m}$	20-30m ² /h
	Rusty layer: $\leq 50\mu\text{m}$	25-50m ² /h

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Laser Cleaning Solutions

Tyre Inner Liner Laser Cleaning Machine

► Automatic work at high efficiency

This self-developed machine is ideally fit to clean silent tyre, self-seal rubber tyre, as well as its inside dead angle by swinging at pitching angle. For tyre, its inner wall surface will not be damaged but surface adhesive force is about to be increased. It can work with other production lines, dock with MES system and identify tyre barcode automatically.

4 Competitive Strengths

- Easy to operate and eco-friendly
- Take precise control and no harm to substrate
- Work efficiently and no worry about high temperature
- Raise frictional coefficient of tyre inner liner



Technical Parameters

Laser power	0.5-1kW
Cleaning mode	Non-contact laser
Cleaning ability	Remove oil, rust, paint and so on
Models we recommend	SF200SHC/SF300SHC/SF500SHC/SF1000SHC

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Laser Cleaning Solutions

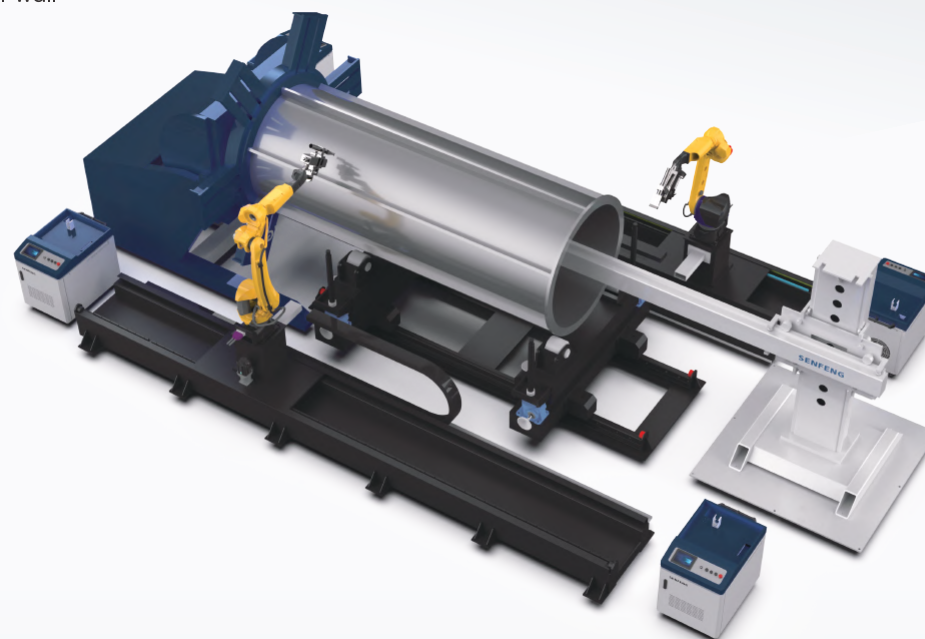
Pressure Vessel Laser Cleaning Machine

► Automatic work at high efficiency

It is applied to cleaning outer walls of LNG interior container, incl. all external wall surfaces, straight arms, two-end seal heads and weld seams.

3 Competitive Strengths

- Easy to operate and eco-friendly
- Take precise control and no harm to substrate
- Work efficiently and no worry about high temperature

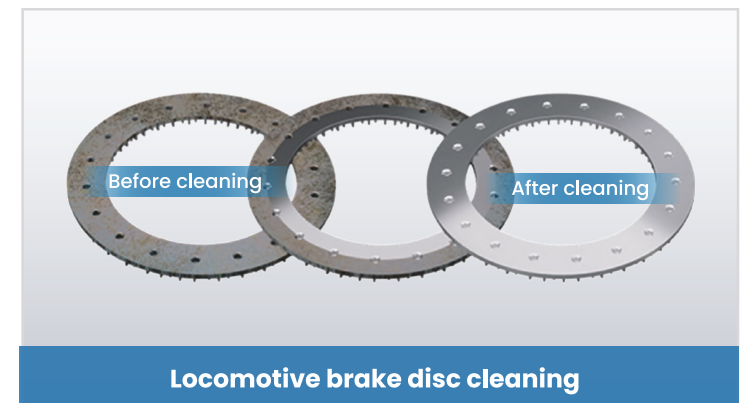
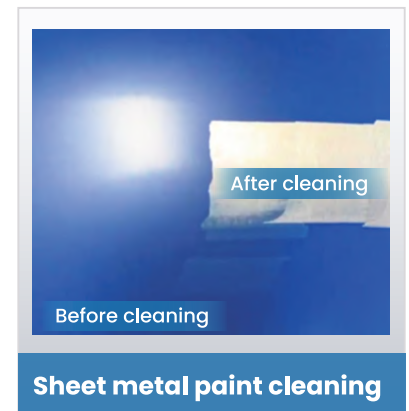
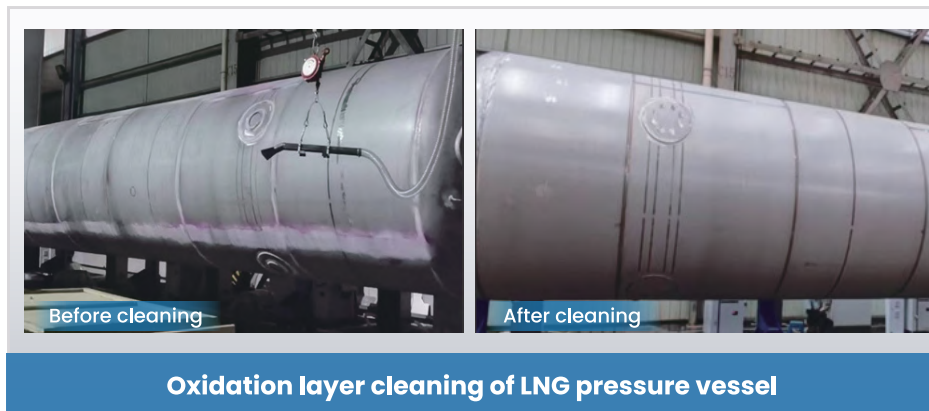


Technical Parameters

Laser power	0.5-1kW
Cleaning mode	Non-contact laser
Cleaning ability	Remove oil, rust, paint and so on
Models we recommend	SF200SHC/SF300SHC/SF500SHC/SF1000SHC

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Laser Cleaning Samples



Applicable Industry



Aerospace



Military industry



Rail transit



Automotive parts



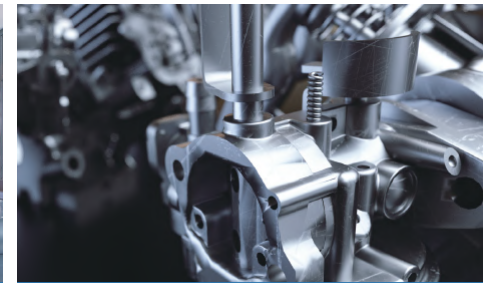
Engineering machinery



New energy



Kitchenware and sanitary ware



Machinery manufacturing



Office furniture



Medical device



Cultural relic repair



Mold production

After-sales Service

Quick-response Service



High Efficiency

Answer your repair calls for 24*7*365 days

Technical engineers can diagnose your machine within 10min, make repair plans within 1h and dispatch to your location within 1 workday after receiving your request for repair



Craftsmanship Spirit

Provide custom-made solutions and make well-targeted plans as the case may be

Engineer certifications: Our all engineers take appointment with certificate after strict training

Train how to fix your FAQ: Collect FAQ and ask the certified engineers to train relevant customers

Online one-to-one guidance: Senior engineers guide customers to get problems solved through phone, video or other online means.

One time is enough: Commission the machine and settle similar matters only once



For Your Peace of Mind

Pre-arrival service: Train in technical theory, actual operation and troubleshooting at great length

Regular service: Reminder of maintenance, door-to-door and sales promotion service

Value-added service: Software & hardware upgrade, finance leasing and further extension of warranty period

After-sales Service





Start VR Tour

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